

Parallel Computing Mini-Project

Team Assignments, Evaluation and Submission Guidelines

1. Team Topics

Team	Theme	Parallel Model	Main Task
1	Parallel Vector Addition	OpenMP	Add large vectors using multiple threads and analyze performance.
2	Parallel Vector Multiplication	OpenMP	Perform element-wise vector multiplication and compare execution time.
3	Parallel Sum and Average of Dataset	OpenMP	Calculate sum/average of a large dataset using multiple threads.
4	Parallel Maximum/Minimum Search	OpenMP	Find maximum and minimum values in a large dataset using threads.
5	Distributed Vector Processing	MPI	Divide a large vector among processes and perform computations.
6	Distributed Dataset Statistics	MPI	Calculate sum, average, maximum and minimum using multiple MPI processes.
7	GPU Vector Operations	CUDA	Perform vector addition/multiplication using CUDA threads and compare CPU/GPU execution.
8	GPU Dataset Processing	CUDA	Process a large numerical dataset using CUDA and analyze execution time.
9	OpenMP vs MPI for Data Processing	OpenMP + MPI	Perform the same simple dataset operation using OpenMP and MPI and compare results.
10	CPU vs GPU Data Processing	OpenMP + CUDA	Perform the same numerical operation using CPU threads and CUDA and compare performance.

2. Checkpoints (5 Checkpoints, 10 Marks)

Checkpoint	Work	Marks
1	Problem definition + sequential algorithm + parallel design	2
2	Working parallel implementation using assigned model	2
3	Run with different data sizes / threads / processes and collect results	2
4	Generate graphs and analyze execution time, speedup and efficiency	2
5	Final demonstration + technical viva	2
	Total	10

3. Submission Guidelines (Experiment 1, Experiment 2 and this Lab Evaluation)

- This applies to **Experiment 1**, **Experiment 2** and **this lab evaluation**. Everything must be uploaded to **GitHub**.
- **One GitHub repository per experiment** (three repositories in total: Experiment 1, Experiment 2 and the lab evaluation).
- **Only one PPT** is to be prepared, for **this lab evaluation** presentation, and it is stored in the lab evaluation repository.
- Every repository must have a proper **README.md** and a clean, organised folder structure.

Suggested repository structure

```
exp-name-parallel-computing/  
|-- README.md          (title, team members, objective, how to build and run, results summary)  
|-- src/               (sequential and parallel source code: .c / .cpp / .cu)  
|-- data/              (input datasets or generator script)  
|-- results/           (raw timing output: csv / txt)  
|-- graphs/            (execution time, speedup, efficiency plots)  
|-- report/            (optional: PDF report)  
`-- presentation/      (the single PPT, lab evaluation repo only)
```